HACKTOBERFEST 26

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Fahes



HACKTOBERFEST PROJECT DEMO

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Green Tech / Circular Economy

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The Problem

E-waste is invisible. A failing laptop, an old router, a drawer of batteries - there is no simple way to decide what to replace, what to recycle, and who will take it. Valuable parts and materials are lost.

Our Solution

Fahes scans this PC, USB devices and the local network, then stashes every item for Recycle or Replace. Replace fetches parts from OpenSooq and Jordanian retailers, then categorises and filters them with the app; Recycle turns photos and quantities into a handover PDF for a verified recycler.

Target Impact

Less e-waste, more reuse and a real paper trail. Every retired item is measured in kilograms diverted and materials recovered; every replacement is matched to the machine. Local-first, offline by default, Arabic and English.

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Open Source AI & Development

AI Integration:

Runtime AI is optional and local: Ollama with a small open-weight model turns a sentence into market filters. Compatibility, prices and impact stay deterministic. We built the app with OpenCode, the open-source coding harness, using DeepSeek through OpenRouter as our development model provider.

No local model? A keyword parser handles the same query, and the UI says which engine answered.

What We Built:

Working Electron app: PC, USB and LAN discovery; recycle/replace stash with edit, notes and photos; AI-assisted OpenSooq search with deterministic compatibility; photo-backed recycle PDF; email handover; impact ledger; Arabic UI; MIT.

3
scan sources: PC / USB / LAN

338
real OpenSooq listings

OpenCode
open-source harness

DeepSeek
via OpenRouter

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Future Implementations

After the hackathon: from filter-filling to part-level intelligence.

Read the part

A local vision + text model reads the model string, label photo or SMART output and extracts the exact part: part number, capacity, DDR generation, DIMM/SO-DIMM, NVMe/SATA, M.2 length, connector and wattage.

Verify fitment

Every OpenSooq listing or retail product is checked against the detected machine before it is shown. Slot, generation, length and power are matched; unknown specs stay 'check'. Confidence is explicit, never guessed.

Rank with confidence

Providers are plug-ins: OpenSooq first, then GTS and local retailers. Normalise price, condition and provenance, rank by confidence and cost, then send the chosen part to the Replace stash.

High confidence
exact part number + fitment verified

Check
one spec missing — ask the seller

Incompatible
known mismatch — hidden by default

Now: local model fills filters • Next: part reader + high-confidence ranking • Then: retail provider adapters + price history

Guardrails: confidence is visible • human confirmation before buying • no invented specs • offline fallback

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Thank You!

Questions & Answers | github.com/SalehGhobari/MLH-Hackathon-_Fahes

OpenCode • DeepSeek via OpenRouter • Electron • MIT

Scan to open the repo

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